Rekindle

Smart glasses that bring faces back.

Simulated smart glasses for recognizing familiar people and daily reminders, with a chatbot for caregivers, built on Microsoft Azure

Project Documentation | Microsoft Machine Learning Track | Digital Egypt Pioneers Initiative (DEPI), Phase Five

Project name	Rekindle
Track	Microsoft Machine Learning Track
Cloud platform	Microsoft Azure
Project type	Educational assistive prototype (graduation project)
Platform	Web application that simulates smart glasses on a phone or laptop, using the device's existing camera
GitHub repository	https://github.com/nayra-ahmedaraby/ReKindle
Team leader	Nayra Ahmed
Submission date	19 October 2026

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Rekindle is an educational assistive prototype. It is not a medical device, does not diagnose, treat, or prevent Alzheimer's disease, and is not a replacement for professional care or caregivers.

PART 1

Project Planning & Management

Idea, objectives, scope, plan, team, risks, and success metrics

1.1 Project Idea

People living with Alzheimer's disease often fail to recognize family members and friends, forget appointments and medication, and ask the same personal questions repeatedly. Rekindle is an assistive system that supports them in these daily situations. The long-term vision is a pair of camera-equipped smart glasses that recognize familiar people and remind the wearer of their daily schedule and medication. Alongside the glasses, an AI chatbot helps family members and caregivers look up the patient's schedule, reminders, and registered people.

In this project the glasses are **simulated**: a web application running on a phone or laptop uses the device's camera as the glasses' camera and shows what the wearer would see and hear. An authorized family member or caregiver manages all familiar-person profiles, approved facts, and schedules and is the only user of the chatbot; the patient-facing view is **display-only** and presents only this approved content. The system is designed, trained, deployed, and monitored on **Microsoft Azure**.

1.2 Objectives

- Train a face-recognition model on free GPU resources (Google Colab / Kaggle), then track, register, and evaluate it with Azure Machine Learning and serve it from Azure Functions so that it identifies caregiver-registered people and reports unknown or uncertain faces instead of guessing.
- Deliver daily schedule and medication reminders to the patient's display-only view at the scheduled time.
- Provide an AI chatbot for family members and caregivers that answers questions about the patient's schedule, reminders, and registered people using records stored in Rekindle only.
- Let caregivers register a familiar person in under a minute with a short guided face video, like phone face unlock, without uploading many photos or retraining the model.
- Give caregivers a secure interface to manage profiles, facts, and reminders.
- Apply Microsoft's Responsible AI principles and protect personal data throughout the system.
- Deliver an integrated MVP deployed on Azure by Week 7 and a tested final system by Week 10, at zero cost to the team.

1.3 Project Scope

The project includes three features:

Feature	Description
Face Recognition (Glasses Simulation)	Simulates the smart glasses' camera using the phone or laptop camera. Recognizes familiar people and shows a green frame with their name and relationship on screen, as the glasses would. A caregiver registers each person with a short guided face video (about 10 seconds), similar to phone face unlock; adding a person does not retrain the model. The model is trained once by the team on free GPUs (Google Colab / Kaggle) and then tracked, registered, and evaluated with Azure Machine Learning, then served from Azure Functions. Unknown or uncertain faces are reported as such rather than guessed.
Daily Schedule & Medication Reminders	Displays the daily schedule and sends reminders triggered by Azure Functions. The system does not verify whether medication was actually taken.
AI Chatbot (for family members and caregivers)	A text chatbot in the caregiver interface, built with Azure OpenAI through Azure AI Foundry, that answers family members' and caregivers' questions about the patient (for example, "Which medications are due today?" or "When is the next appointment?") using only the records stored in Rekindle. If information is missing, it says so instead of inventing an answer. The patient does not use the chatbot.

- **Platform:** A web application, hosted on Azure, that runs on a phone or laptop and uses its built-in camera to simulate the glasses.
- **Users:** The patient view is display-only: it shows recognized names, today's schedule, and reminders, and accepts no typed input. Family members and caregivers use the caregiver interface and the chatbot.
- **Hardware:** No hardware is built in this project. Physical smart glasses are future work.
- **Future work:** Physical smart glasses, and a voice assistant that will become the patient's input channel (asking questions and hearing answers and announcements hands-free).
- **Out of scope:** Fall detection, medical diagnosis, treatment advice, and proof of medication intake.
- **Status:** Planning / early development. No feature has been implemented yet and no results are reported; Azure tiers and model choices remain subject to feasibility testing.

1.4 Microsoft Azure Technology Stack

Rekindle's machine-learning lifecycle and cloud services are built on Microsoft Azure [15]. The project has **no budget**: every component uses a free service, a free tier, or the Azure for Students credit, which needs no credit card, and no paid compute runs continuously.

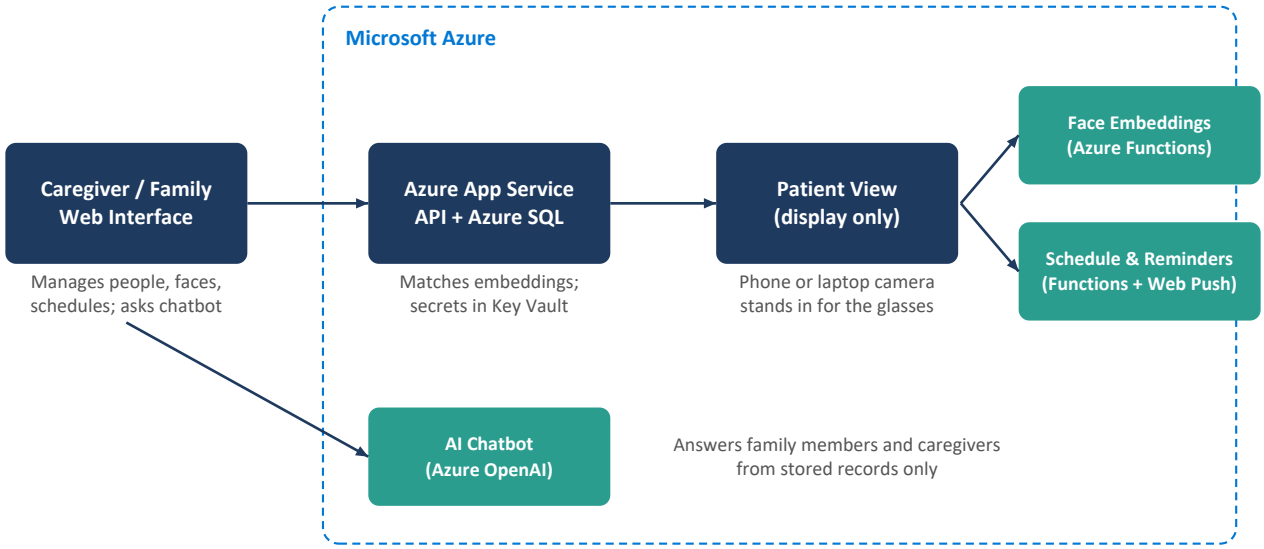
Component	Service / tool	Purpose	How it stays free
Face detection	MediaPipe Face Detector (BlazeFace), in the browser [17]	Detect and crop faces on the device; only the face crop is sent to Azure.	Open source; runs on the user's device.
Model training	Google Colab / Kaggle GPUs	Train or fine-tune the embedding model once.	Free GPU quotas.
ML lifecycle	Azure Machine Learning: MLflow tracking, model registry, Responsible AI dashboard	Track runs from Colab / Kaggle; version and evaluate the model.	No charge for the workspace; evaluation jobs use a cluster that scales to zero, within the student credit.
Embedding service	Azure Functions (Python, ONNX Runtime) with the registered model	Return the embedding of a face crop; used for both enrollment and recognition.	Monthly free grant of Azure Functions.
Matching & backend	Azure App Service (FastAPI), Azure SQL Database, Blob Storage	Compare the returned embedding with stored embeddings (cosine similarity + threshold); host the API and web views; store profiles, embeddings, and thumbnails.	App Service free tier (F1), Azure SQL Database free offer, small storage within the student credit.
Reminders	Azure Functions timer triggers + Web Push with VAPID keys [18]	Push reminders to the patient's browser.	Functions free grant; Web Push itself is free.
Caregiver chatbot	Azure OpenAI (e.g., GPT-4o mini) through Azure AI Foundry + Azure AI Content Safety	Answer caregivers from stored records only; filter unsafe content.	Low token use within the student credit; Content Safety free tier (F0); GitHub Models free tier as fallback.
Security & operations	Microsoft Entra ID, Key Vault, Application Insights, GitHub Actions	Caregiver sign-in, secret storage, monitoring, and automated deployment.	Entra ID Free, Application Insights free monthly data, GitHub Actions free for public repositories.

Training and serving: the team trains or fine-tunes the face-recognition model (FaceNet / ArcFace embeddings) once, on free GPUs in Google Colab or Kaggle. Each run logs its parameters, metrics, and artifacts to the Azure ML workspace through MLflow; the chosen model is registered in the Azure ML model registry, evaluated with the Responsible AI dashboard, exported to ONNX, and served by Azure Functions. The embedding service only returns an embedding; the backend on App Service compares it with the stored embeddings using cosine similarity and the threshold, so the same service is used for enrollment and recognition and never needs database access. Families

never train the model: a new person is simply a new set of stored embeddings. An always-on Azure ML managed online endpoint is not used because it is billed per hour even when idle.

Note: the Azure AI Face service's identification features are Limited Access and require Microsoft approval. The team therefore trains its own face-recognition model, manages it in Azure Machine Learning, and will use Azure AI Face only if access is granted.

1.5 System Workflow



The patient view is display-only: it shows who is in front of the patient and today's reminders, with no typed input. Family members and caregivers register familiar people with a guided face video, manage approved facts and reminders, and use the chatbot to look up the patient's records. All data and models stay inside the team's Azure subscription.

1.6 Timeline

This is a proposed timeline, not an official DEPI schedule.

Week	Activities	Deliverable
Week 1	Confirm scope and requirements, create the GitHub repository and Azure resource group, assign responsibilities.	Repository, task board
Week 2	Design the user interface, database, API, and Azure architecture; choose the training dataset; set up the Azure ML workspace.	Design documents
Week 3	Prepare datasets and module prototypes; run first training experiments on Colab / Kaggle GPUs, logged to Azure ML with MLflow.	First model baseline
Week 4	Implement face recognition, the guided enrollment UI, reminders, the chatbot, and the glasses-simulation view in parallel.	Working modules
Week 5	Continue implementation, deploy the embedding service on Azure Functions, and test individual modules.	Embedding service
Week 6	Integrate the three features on Azure.	Integrated build
Week 7	Deliver an integrated MVP deployed on Azure.	MVP
Week 8	Test the complete system, review the Responsible AI dashboard, and fix errors.	Test report
Week 9	Improve the interface; prepare documentation, README, and presentation slides.	Final documentation
Week 10	Final review, demo preparation, and submission.	Final demo

1.7 Team Members and Responsibilities

All members test their own modules and contribute to integration on Azure and the final presentation.

#	Member	Role	Responsibilities
1	Nayra Ahmed	Team Leader & Integration	Coordinate tasks and deadlines, manage GitHub, documentation, and the Azure subscription and resource groups, track requirements and blockers, support API integration, and coordinate integration testing.
2	Nada Kotkat	Face Recognition	Train and test the face-recognition model on Colab / Kaggle GPUs, track it with MLflow in Azure ML, register and evaluate it, and deploy the embedding service on Azure Functions.
3	Mariam Mohamed	AI Chatbot	Develop the caregiver-facing chatbot with Azure OpenAI (Azure AI Foundry) over the records stored in Rekindle, and support face-recognition data preparation and testing.
4	Michael Samaan	Reminders & Daily Schedule	Develop the scheduling and reminder module with Azure Functions.
5	Naheel Mohamed	Backend & Database	Handle backend APIs on Azure App Service, Azure SQL database design, Blob Storage, and data management.
6	Mahmoud Elsoly	Frontend	Build the caregiver interface (including the chatbot and the guided enrollment video with its green guide ring) and the display-only glasses-simulation view with its green and grey face frames, connect them to the backend, and deploy on Azure App Service.

1.8 Risks and Mitigation

Risk	Impact	Mitigation
False face identification	High	Conservative threshold, unknown-face handling, testing under varied lighting and angles, evaluation in Azure ML.
Poor-quality enrollment video	Medium	The guide ring turns green only for clear, well-lit frames at each angle; the caregiver is asked to retake missing angles.
Simulation differs from real glasses	Medium	Document the differences (camera angle, field of view, processing power); treat hardware as future work.
Chatbot gives family members wrong information	High	Answers only from stored records, with the source record shown; fallback for missing facts; content filtering.
Reminders fail in background or offline	Medium	Test browser and device states; document limitations.
Exposure of personal data	High	Consent, Entra ID access control, private Blob Storage, Key Vault, unregistered faces never stored, compliance with Egypt's Law No. 151 of 2020 [19], no secrets in the repository.
Azure credits run out or access is denied	High	Free tiers first, no always-on paid compute (no managed online endpoint), budget alert on the student credit, compute that scales to zero, train on free Colab / Kaggle GPUs, self-hosted model instead of Limited Access services.
Azure OpenAI not available in the subscription	Medium	Use another model from the Azure AI Foundry model catalog, or the GitHub Models free tier.
Free GPU sessions are limited or interrupted	Medium	Save checkpoints regularly, keep runs short, log every run to Azure ML with MLflow, and switch between Colab and Kaggle.
Scope exceeds available time	Medium	Fixed three-feature scope; complete one end-to-end flow on Azure first.

1.9 Success Metrics (KPIs)

The following targets will be measured on consented or synthetic test data during Week 8. They are goals, not results.

Area	Metric	Target
Face recognition	Correct identification of registered people (test set)	≥ 90%
Face recognition	False matches (wrong name shown) at the chosen threshold	≤ 2%
Face recognition	Unregistered faces correctly reported as “unknown”	≥ 95%
Face recognition	Time from camera frame to on-screen result	≤ 2 seconds
Reminders	Reminders delivered within 1 minute of the scheduled time (app open)	≥ 95%
Chatbot	Answers supported by a stored record, with the source shown (fixed set of caregiver test questions)	≥ 95%
Chatbot	Fallback message shown when the record is missing (caregiver test questions)	≥ 95%
Usability	System Usability Scale (SUS) score from test users	≥ 70
Project	Integrated MVP deployed on Azure	By Week 7

PART 2

Literature Review

Similar solutions, comparison, and lessons applied to Rekindle

2.1 Background

According to the World Health Organization, around 57 million people were living with dementia worldwide in 2021, with nearly 10 million new cases every year; Alzheimer's disease may contribute to 60–70% of cases [1]. Dementia organizations recommend memory aids such as calendars, reminders, labelled photos, and consistent daily routines [2][3], and a randomized controlled trial found that smartphone reminder tools improved prospective memory in older adults [4]. These sources support the need for Rekindle; they do not show that Rekindle itself is clinically effective.

2.2 Similar Solutions

Solution	Type	Strengths	Limitations for our users
OrCam MyEye [6]	Wearable camera that attaches to glasses	Works offline; reads text and recognizes saved faces in real time.	Designed for blind and low-vision users, not dementia; high cost; no reminders or caregiver tools.
Envision Glasses [7]	Smart glasses with an AI assistant	Hands-free; describes scenes, reads text, and can recognize faces taught by the user.	Designed for blind and low-vision users; high cost; no caregiver-managed schedule.
Microsoft Seeing AI [8]	Free mobile app	Free; uses the phone camera; describes people and scenes and reads text.	Designed for blind and low-vision users; no reminders, caregiver control, or dementia focus.
Microsoft Project Tokyo [9]	Research prototype on a modified HoloLens	Showed that head-worn cameras can tell a user who is nearby, with audio cues.	Research only; specialized hardware; not available as a product.
Ray-Ban Meta smart glasses [10]	Consumer camera glasses with an AI assistant	Affordable mainstream camera glasses; voice assistant; photo and video capture.	No face recognition of familiar people; general-purpose, not designed for dementia.
Medisafe [11]	Medication reminder app	Reliable medication reminders; caregiver notifications for missed doses.	No face recognition; the patient must operate the app.
Voice assistants (Amazon Alexa, Google Assistant)	Smart speakers and phone assistants	Hands-free reminders and general questions.	General web answers can be wrong or confusing; no face recognition; not limited to approved information.

2.3 Comparison with Rekindle

Capability	OrCam	Envision	Seeing AI	Medisafe	Rekindle
Recognizes familiar people	Yes	Yes	Partial	No	Yes
Daily schedule & medication reminders	No	No	No	Yes	Yes
Chatbot for caregivers over the patient's records	No	No	No	No	Yes
Caregiver manages the content	No	No	No	Partial	Yes
Designed for people with dementia	No	No	No	No	Yes
Runs on a device the user already owns	No	No	Yes	Yes	Yes

Existing tools each solve one part of the problem. Rekindle combines face recognition and reminders for the patient, in a display-only view, with a chatbot that helps family members look up the patient's records, in one caregiver-controlled system designed for people with dementia that runs on a phone or laptop the family already owns.

2.4 Technical Approaches Reviewed

- **Face embeddings:** FaceNet maps faces to compact embeddings so that faces of the same person are close together, enabling recognition from a few reference images, so new people can be added without retraining [5]. ArcFace improves the separation between identities with an additive angular margin loss [12]. Rekindle will train both on Colab / Kaggle GPUs, compare them in Azure ML, and choose based on accuracy and speed.
- **Face detection:** BlazeFace is a lightweight detector built for mobile GPUs [17]; running it in the browser keeps full camera frames on the device. Models are benchmarked on Labeled Faces in the Wild (LFW) [16].
- **Threshold-based matching:** Comparing embedding distance against a threshold lets the system answer "unknown" instead of forcing a match, which is essential to avoid naming the wrong person.
- **Grounded chatbot answers:** Retrieval-augmented generation supplies the language model with retrieved records so that answers are based on supplied facts rather than the model's general knowledge [13]. Rekindle applies this idea so that answers to family members and caregivers come only from records stored in the system.
- **Responsible AI:** Microsoft's Responsible AI principles guide fairness testing, privacy, and transparency [14].

2.5 Lessons Applied to Rekindle

- Keep the experience hands-free and glanceable, as wearable assistants do, by designing the patient view as a glasses display.
- Use existing devices first to keep costs low, and treat dedicated glasses as future work.
- Borrow the guided enrollment pattern of phone face unlock so that families can register a person in seconds.
- Prefer "I'm not sure" over a wrong name or a wrong answer.
- Keep the patient's interaction passive (display only) until hands-free voice input is available.
- Put caregivers in control of all personal information, as medication apps do with caregiver notifications.

PART 3

Requirements Gathering

Stakeholders, functional, quality, and data requirements

3.1 Stakeholders and Their Needs

Stakeholder	Role	Needs
Patient (person living with Alzheimer's)	Primary user (view only)	Know who is in front of them and what is next in the day without typing or navigating; receive calm, clear, consistent reminders.
Caregiver / family member	Content administrator and chatbot user	Register a familiar person quickly with one short video, without uploading many photos; manage schedules, reminders, and approved facts; quickly look up the patient's records through the chatbot; trust that the system will not show wrong information; keep the patient's data private.
Familiar people (relatives, friends)	People being recognized	Give consent before their face is enrolled; be able to have their face data deleted; people who are not registered are never stored.
Healthcare professionals	Indirect stakeholder	The system must not give medical advice or replace professional care.
DEPI evaluators and instructors	Project assessors	Clear use of machine learning and Microsoft Azure, documented requirements, measurable results, and a working demo.
Development team	Builders and maintainers	Clear requirements, modular architecture, and a maintainable codebase in one GitHub repository.

3.2 Functional Requirements

ID	Requirement
Caregiver & account management	
FR-01	The caregiver can sign in securely using Microsoft Entra ID.
FR-02	The caregiver can create, edit, and delete familiar-person profiles (name, relationship, short note).
FR-03	The caregiver can enroll a familiar person, with their consent, by recording a guided face video of about 10 seconds; a guide ring turns green as each angle (front, left, right, up, down) is captured.
FR-04	The caregiver can add, edit, and deactivate approved facts used by the chatbot.
Face recognition (glasses simulation)	
FR-05	The caregiver opens the patient view on the patient's device and grants camera and notification permissions once during setup.
FR-06	Enrollment selects 5–10 clear frames from the video, computes their embeddings, and stores only the embeddings and one profile thumbnail; the video is discarded and the model is not retrained.
FR-07	Faces are detected in the browser with MediaPipe Face Detector; only the face crop is sent over HTTPS to the embedding service, and the backend compares the returned embedding with the stored embeddings (cosine similarity and threshold).
FR-08	When a match is above the threshold, a green frame appears around the face with the person's name and relationship as a glasses-style overlay.
FR-09	When no match is above the threshold, a grey frame appears with "I'm not sure who this is" instead of a name.

Daily schedule & reminders	
FR-10	The caregiver can create one-time and recurring events and medication reminders with a time and short message.
FR-11	The patient view shows today's schedule in time order.
FR-12	The system sends a Web Push notification with sound at the scheduled time while notifications are permitted.
FR-13	Reminders are shown full-screen on the patient view and clear automatically after a set time; the system does not record or claim that medication was taken.
AI chatbot (family members and caregivers)	
FR-14	A signed-in family member or caregiver can type a question to the chatbot in the caregiver interface.
FR-15	The chatbot answers using only the latest active records stored in Rekindle and shows the record each answer is based on.
FR-16	If no stored record answers the question, the chatbot replies that it does not have that information.
FR-17	The chatbot refuses medical-advice questions and suggests contacting a doctor.
Patient view	
FR-18	The patient view is display-only: it shows recognized names, today's schedule, and active reminders, and has no text input or chatbot.

3.3 Quality (Non-Functional) Requirements

ID	Category	Requirement
NFR-01	Performance	Face-recognition result is shown within 2 seconds of the face appearing in the camera view.
NFR-02	Performance	Chatbot replies to a caregiver within 5 seconds for a typical question.
NFR-03	Accuracy	At the chosen threshold, wrong-name results are 2% or fewer on the test set; uncertain cases show "unknown".
NFR-04	Reliability	Reminders are delivered within 1 minute of the scheduled time in tested browser states; known limitations are documented.
NFR-05	Security	Caregiver functions require authentication; the patient view is read-only.
NFR-06	Security	All traffic uses HTTPS; keys and connection strings are stored in Azure Key Vault, never in the repository.
NFR-07	Privacy	Only face embeddings and one profile thumbnail per consenting registered person are stored, in private storage; enrollment videos and camera frames are discarded after processing; unregistered faces are never stored; data can be deleted on request.
NFR-08	Privacy	Development and testing use synthetic or consented data; no real patient data is stored in GitHub.
NFR-09	Usability	The patient view needs no typing or navigation; it uses large readable text, high contrast, and calm, consistent wording.
NFR-10	Usability	The patient view works on common phone and laptop browsers at phone screen width.
NFR-11	Responsible AI	Model errors are reviewed with the Azure ML Responsible AI dashboard; chatbot inputs and outputs pass Azure AI Content Safety.
NFR-12	Maintainability	Modules (face recognition, reminders, chatbot, backend, frontend) are separated and deployed through GitHub Actions.

ID	Category	Requirement
NFR-13	Cost	Zero cost to the team: training uses free Colab / Kaggle GPUs; the deployed system uses only free services, free tiers, and the Azure for Students credit (no credit card); no always-on paid compute; a budget alert is set on the credit.
NFR-14	Compliance	Personal data, including biometric and health-related data, is processed in line with Egypt's Personal Data Protection Law No. 151 of 2020 [19]: explicit consent, purpose limitation, minimum data, and deletion on request.

3.4 Data Requirements

No real patient data is used. The training dataset has not been chosen yet; the team will select it in Week 2 after checking licences.

Data	Source	Use
Model training data	A public face dataset (for example from Kaggle, such as LFW [16]) or public pretrained FaceNet / ArcFace weights, or data collected by the team; to be decided in Week 2	Used once by the team to train or fine-tune the model on Colab / Kaggle GPUs. Families never train the model.
Enrollment data	One guided face video (about 10 seconds) per familiar person, recorded by the caregiver with consent	5–10 clear frames are turned into embeddings; the video is then discarded.
Evaluation data	Team members and volunteers who give written consent, enrolled through the same video flow, plus public images of people who are not registered	Measures the face-recognition KPIs; the test-set size will be fixed once the dataset is chosen.
Chatbot and reminder data	Synthetic patient profile created by the team	Synthetic schedules, reminders, and facts, with caregiver test questions, including questions about missing information.

- Adding a person needs no training: the model turns faces into embeddings, and a new person is simply a new set of stored embeddings.
- The recognition threshold is tuned on validation images, and the KPIs are measured on separate test images.
- Camera frames are processed in memory and discarded; faces of unregistered people are never stored, and logs keep only the result label and confidence, never images.
- Deleting a profile removes its embeddings and thumbnail.

3.5 Constraints and Assumptions

- The system runs as a web application on a phone or laptop; no glasses or other hardware are built.
- Deployment, experiment tracking, model registry, and evaluation run on Microsoft Azure, as required by the Microsoft Machine Learning Track; only GPU training runs on Google Colab / Kaggle.
- Users have an internet connection; recognition and the chatbot require access to Azure.
- Browser notification behaviour varies by device; delivery when the app is closed is not guaranteed, and on iPhone Web Push works only after the web app is added to the home screen.
- The project has no budget: no paid service or subscription is used.
- The project must be completed within the 10-week timeline by a team of six.

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